

WBM Interview Questions:

WBM is Windchill Bulk Migration.

Migration: Migration constitutes of migrating data into Windchill.

1. Please describe the WBM Utility.

Ans:

- The Windchill Bulk Migrator is a collection of tools which are meant to facilitate the permanent, one-way migration of data into Windchill.
- Use the Windchill Bulk Migrator to extract data from Windchill source systems, and load the standardized data from the staging database into the target Windchill system.
- There are 3 types of Migrations:

Source

WC2WC

Non WC2WC/Legacy Application

File server/Flat file (FS2WC)

Target

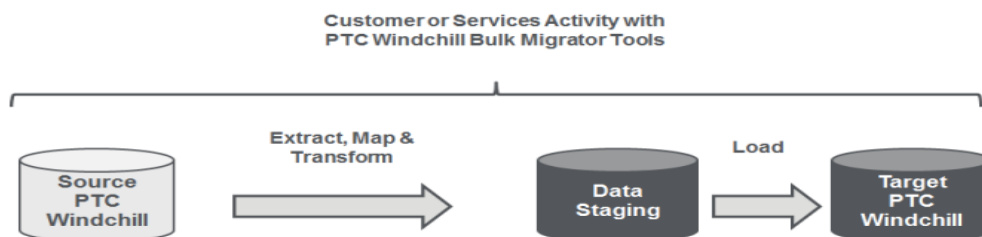
Windchill

Windchill

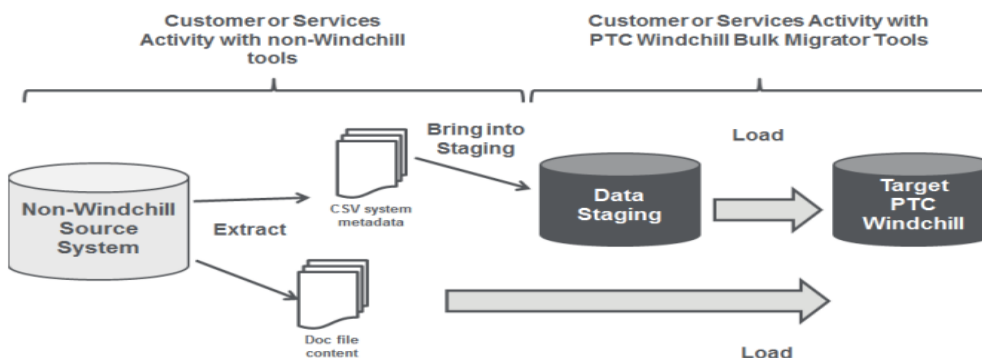
Windchill

The following diagram provides an overview of the process

Windchill-to-Windchill Migration Example



Non-Windchill System to PTC Windchill Migration Example



Migration process have 4 steps as follows,

E(Extraction): WBM will handle.

T(Transformation): WBM will handle.

L>Loading): WBM will handle.

V(Validation): Write own SQL script or using Windchill UI

This ETLV migrations can be used in the following situations:

- When a business wants to select specific data sets to move or split off.
- When the source system is very different from the target system (implying that data mapping and transformation is needed).
- When data from two or more separate sources needs to be combined.

So below are the Migration process includes ETLV,

1. WC2WC:

- a. Source will be WC.
 - b. Target system is WC.
-
- E(Extraction): WBM will handle.
 - T(Transformation): WBM will handle.
 - L>Loading): WBM will handle.
 - V(Validation): Write own SQL script or using Windchill UI.

2. Non WC2WC:

- a. Source will be some other application.
 - b. Target system is WC.
-
- E(Extraction): WBM cannot handle.
 - T(Transformation): WBM will handle.
 - L>Loading): WBM will handle.
 - V(Validation): Write own SQL script or using Windchill UI.

3. File server to WC:

- a. Source is File Directory.
 - b. Target is Windchill.
-
- E(Extraction): WBMFF i.e. Windchill Bulk Migrator from files utility will help.
 - T(Transformation): WBM will handle (SQL Scripts to perform expected transformation).
 - L>Loading): WBM will handle.
 - V(Validation): SQL scripts.

2. WBM installation overview and how to choose the right version of WBM.

Ans:

Windchill 11.0 is Source,

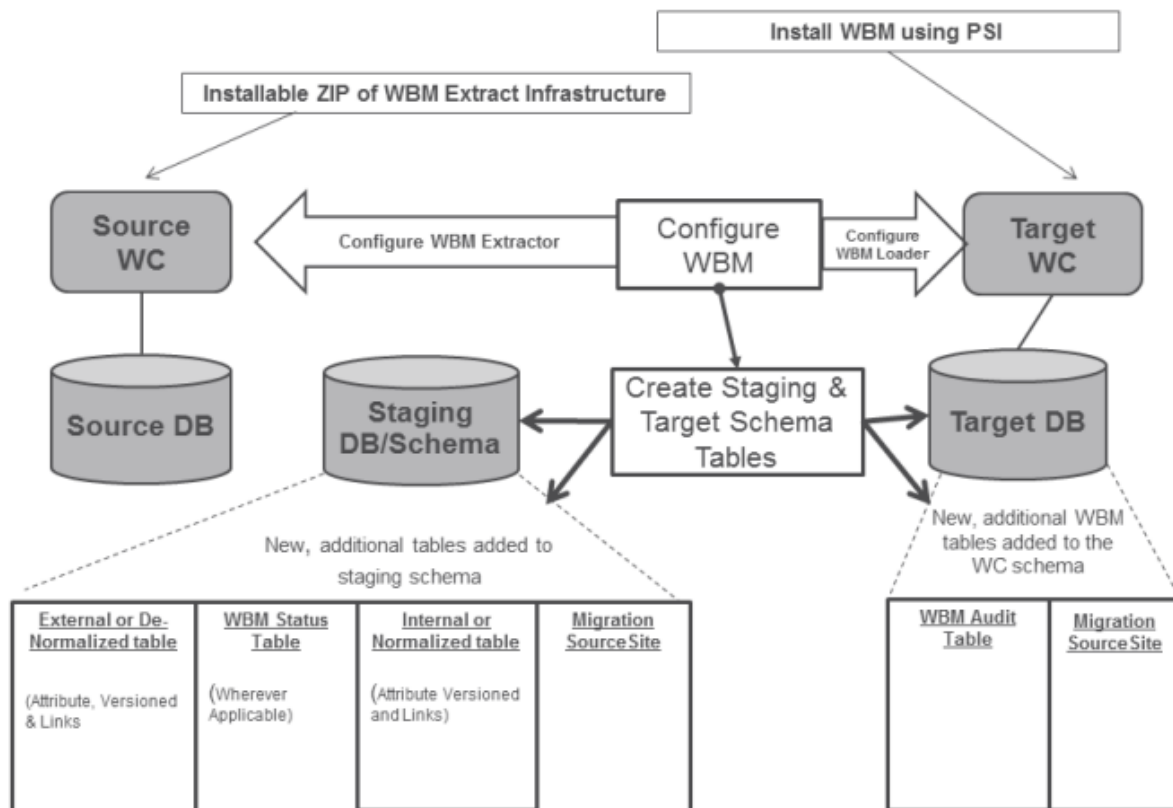
Windchill 12.0 is Target.

Source Windchill version:

M030-CPS11 11.0 pdml.11.0.30.11.29 01 windchill pdmlink 11.0

WBM Installation Overview:

The following image shows a graphical representation of the installation process:



Below are the general steps before configuring and using the Windchill Bulk Migrator:

1. Install Windchill Bulk Migrator.
2. Install the Windchill Bulk Migrator Extraction Utilities.
3. Generate the staging schema.

- Install Windchill Bulk Migrator using Windchill PSI as an optional module.
- Verify the installation by running the Windchill version command.
- Proceed to the installation of the staging schema.

Below version is matching with the Source Windchill version as above

Release	Source Windchill PDM/PJL	Target Windchill PDM/PJL
WBM 11.0 M030 CPS02	10.1 F000+ 10.2 M030+ 11.0 M010+	11.0 M030 CPS08+
WBM 11.0 M020 CPS01	10.1 F000+ 10.2 M030+ 11.0 M010+	11.0 M020 CPS03+
WBM 11.0 M020 F000	10.1 F000+ 10.2 M030+ 11.0 M010+	11.0 M020 CPS02
WBM 11.0 M010	10.1 F000+ 10.2 M030+ 11.0 M010	11.0 M010
4.0 M020	9.1 F000+ 10.0 F000+ 10.1 F000+ 10.2 F000+	10.2 M030 10.2 M032
4.0 M010	9.1 F000+ 10.0 F000+ 10.1 F000+ 10.2 F000+	10.2 M010 10.2 M020 10.2 M022
2.2 M030	9.1 F000+ 10.0 F000+	10.0 M020+ 10.1 F000+

The Windchill Bulk Migrator is installed on the same system where the target Windchill 11.x system is installed. The components are installed into the Windchill 11.x codebase.

The Windchill Bulk Migrator loads data into the target Windchill system from a staging area. The staging area requires a database installation and schema instance devoted to the Windchill Bulk Migration staging schema. The tools read data from the staging database and then create objects on the target Windchill 11.x system and database instance.

The following tables describe the database versions that are supported for the Windchill Bulk Migrator schema and the target Windchill system.

WBM Release	Source Windchill Database Application		Windchill Bulk Migrator Staging Database Application		Target Windchill Database Application	
WBM 11.0	Oracle 11.2.0.3	SQLServer 2008 R2	Oracle 12.1.0.2.0	SQLServer 2012	Oracle 12.1.0.2.0	SQLServer 2012
	M030 CPS02	SQLServer 2012	Oracle 12.2.0.1.0	SQLServer 2014	Oracle 12.2.0.1.0	SQLServer 2014
	Oracle 11.2.0.4	SQLServer 2014		SQLServer 2016		SQLServer 2016
	Oracle 12.1.0.2.0					
	Oracle 12.2.0.1.0					
WBM 11.0	Oracle 11.2.0.3	SQLServer 2012	Oracle 11.2.0.4	SQLServer 2012	Oracle 11.2.0.4	SQLServer 2012
M020	Oracle 11.2.0.4	SQLServer 2008 R2	Oracle 12.1.0.2.0	SQLServer 2014	Oracle 12.1.0.2.0	SQLServer 2014
	Oracle 12.1.0.2.0			SQLServer 2016		SQLServer 2016
WBM 11.0	Oracle 11.2.0.3	SQLServer 2012	Oracle 11.2.0.4	SQLServer 2012	Oracle 11.2.0.4	SQLServer 2012
M010	Oracle 11.2.0.4	SQLServer 2008 R2	Oracle 12.1.0.2.0		Oracle 12.1.0.2.0	
	Oracle 12.1.0.2.0					
4.0 M020	Oracle 11.2.0.3	SQLServer 2008 R2	Oracle 11.2.0.4	SQLServer 2008 R2	Oracle 11.2.0.4	SQLServer 2008 R2
	Oracle 11.2.0.4	SQLServer 2012	Oracle 12.1.0.2.0	SQLServer 2012	Oracle 12.1.0.2.0	SQLServer 2012

	Oracle 12.1.0.2.0					
4.0 M010	Oracle 11.2.0.3	SQLServer 2008 R2	Oracle 11.2.0.3	SQLServer 2008 R2	Oracle 11.2.0.3	SQLServer 2008 R2
2.2 M030	Oracle 11.2.0.3	SQLServer 2008 R2	Oracle 11.2.0.3	SQLServer 2008 R2	Oracle 11.2.0.3	SQLServer 2008 R2

3. How do we configure WBM after the installation.

Ans:

Configuration of WBM:

- Configure the staging database, i.e Staging Schema.
- Configure the type of migration, i.e wc2wc, or nonwc2wc, or file based to wc.

Command use as: `ant -f bin\configure_WBM.xml`. using this command because I have open the command from home folder Windchill.

Please note: if you have choose directory path as windchill\bin then add directly as `configure_WBM.xml`.

* To complete the configuration of the Windchill Bulk Migrator on the target system, complete the following steps:

Target System:

☐ Pre-requisites

- Target Windchill system installed and available
- Windchill Bulk Migrator should be installed on target system using PSI.
- Staging database created and available.
- Documented list of source and target type and attribute mappings

☐ Configure Windchill Bulk Migrator by running ant script as follows:

Step 1. Run ANT installer from a Windchill shell:

- On Windows cmd, enter the following: `ant -f bin\configure_WBM.xml`

Go to Oracle SQL developer -> create a User Staging database as follows:

R/C on System -> Click on Connect -> create a user by giving command as below,

Command: Create user staging identified by staging; -> run the query by CTRL enter.

grant all privileged to staging; -> run the query by CTRL Enter.

Click on New Connection (+) at the top -> Give Connection Name : Staging, User name & Password is same as "staging" -> save password.

Host Name : localhost or 127.0.0.1 & add SID : wind -> click on test -> status shows success -> click on connect.

Step 2. After running, the -ant script prompts you for information regarding the target Windchill

system and the staging system.

The following output shows the installation prompts and input values entered (formatted to fit on the page):

Oracle [input] Enter valid staging database instance URL {e.g. i4605.ptcnet.ptc.com: 1521:SID} in cmd host:1521: wind-> enter.

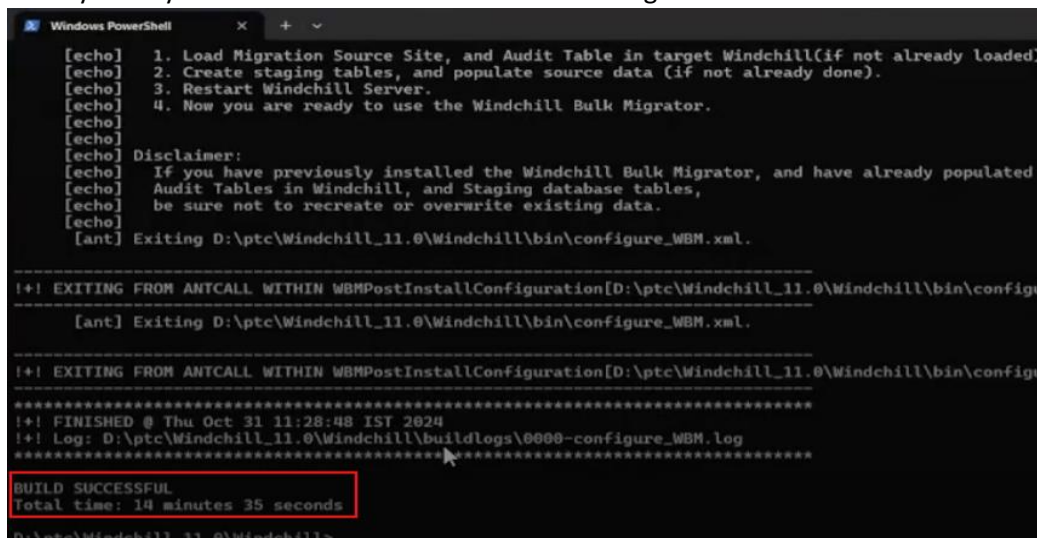
Enter valid staging database user: staging

Enter valid staging database password: staging password

Re-enter valid staging database password: staging password.

Enter staging database type (Oracle|SQLServer): Oracle.

Enter yes Do you want to install the Windchill Bulk Migrator? Y -> enter.



```
Windows PowerShell
[echo] 1. Load Migration Source Site, and Audit Table in target Windchill(if not already loaded)
[echo] 2. Create staging tables, and populate source data (if not already done).
[echo] 3. Restart Windchill Server.
[echo] 4. Now you are ready to use the Windchill Bulk Migrator.
[echo]
[echo] Disclaimer:
[echo] If you have previously installed the Windchill Bulk Migrator, and have already populated
[echo] Audit Tables in Windchill, and Staging database tables,
[echo] be sure not to recreate or overwrite existing data.
[echo]
[ant] Exiting D:\ptc\Windchill_11.0\Windchill\bin\configure_WBM.xml.

[+] EXITING FROM ANTCALL WITHIN WBMPostInstallConfiguration[D:\ptc\Windchill_11.0\Windchill\bin\configu
[ant] Exiting D:\ptc\Windchill_11.0\Windchill\bin\configure_WBM.xml.

[+] EXITING FROM ANTCALL WITHIN WBMPostInstallConfiguration[D:\ptc\Windchill_11.0\Windchill\bin\configu
*****
[+] FINISHED @ Thu Oct 31 11:28:48 IST 2024
[+] Log: D:\ptc\Windchill_11.0\Windchill\buildlogs\0000-configure_WBM.log
*****
BUILD SUCCESSFUL
Total time: 14 minutes 35 seconds
D:\ptc\Windchill_11.0\Windchill>
```

Then the Build successful statement comes shows configuration of WBM done.

4. What is the use of the wbmgen command?

Ans:

The purpose of using the wbmgen command are as follows:

- Configure the type of Migration, i.e. wc2wc, or non wc2wc, or filebased to wc.
- Create some .sql files – for creating tables in staging database.

After running the wbmgen command we can see Start generate AllWormXMLS.

Wbmgen command is going to create .sql files,

Note: 1. If the type of migration is non wc to wc, then wbmgen command creates additional .sql files.

2. If the type of migration is wc to wc, then a little less number of .sql files are created.

4. How do we create a staging schema?

Ans: To create Staging Schema:

Always choose 2. Option i.e. Generate WBM staging schema for a non windchill source system -> enter.

Then table files get created & store it in database.

Go to windchill home directory -> open db folder -> WBM has created SQL files as postgresSQL, sql, sql3, sqlServer.

Below steps to check the tables in groups .sql files,

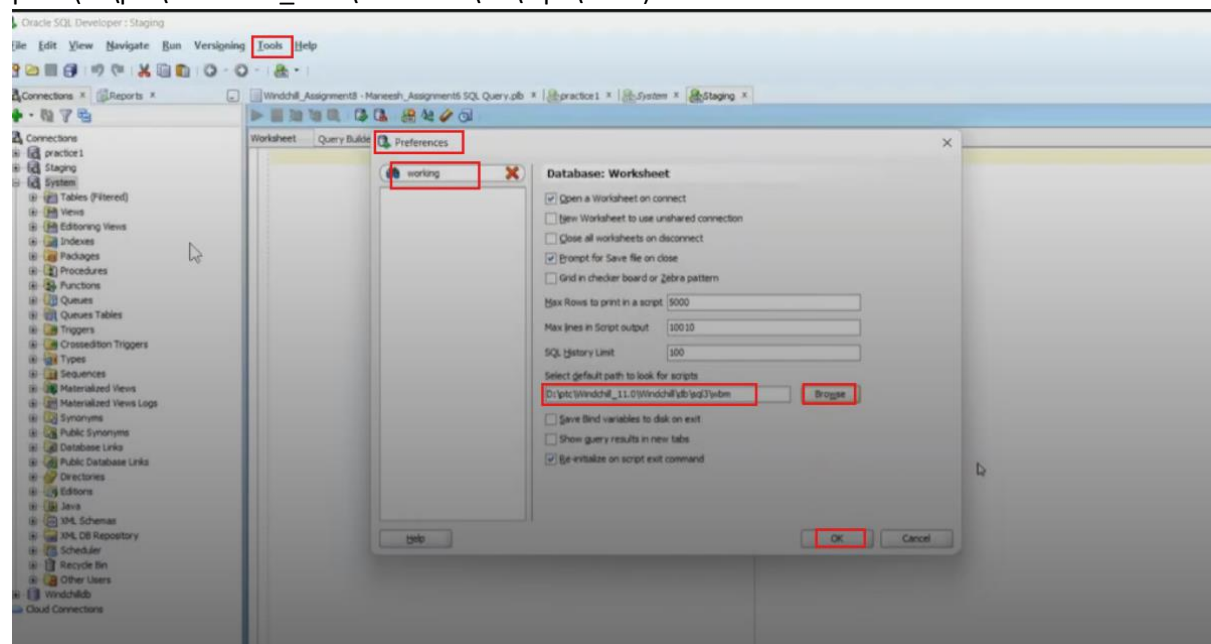
Step1: Open sql3 folder -> we can see create_user.sql -> open in Notepad++.

Step 2: Open sql3 -> open wbm -> we can see create_staging_schema.sql -> open in Notepad++.

Step 3: In sql3-> wbm -> Open stage folder -> we can see create_wtpartsubstitutelink.sql -> open in Notepad++.

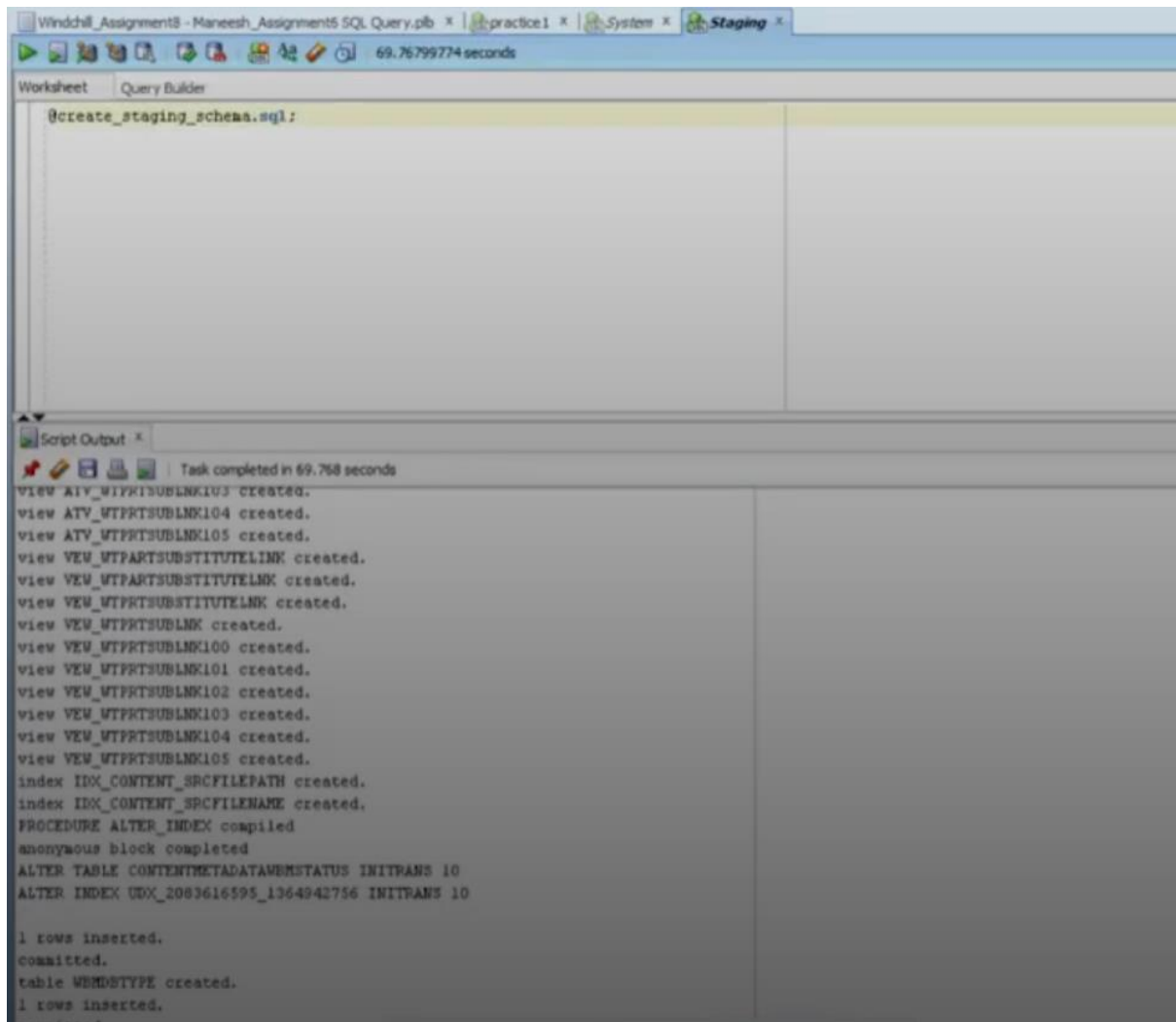
To create all the tables we have to specify the working directory in worksheet to call that create staging schema .sql files.

Step 1: Updating directory -> Go to Tools -> go to preference -> search working directory -> Database: Worksheet box open -> Under select default path to look for scripts -> click on browse give a correct path (D:\ptc\Windchill_11.0\Windchill\db\sql3\wbm).



Step 2: Now call staging schema file by typing @create_staging_schema.sql; -> CTRL enter.

You will find the staging schema tables in sql output.



Note: To creating staging schema alone, it took approx. 3 hours.

6. What is the staging schema and how do we create the it?

Ans:

The staging database is where object metadata is stored prior to being loaded into Windchill. The Windchill Bulk Migrator staging schema is the neutral data format for mapping data to the target Windchill system.

Staging schema represents a neutral format for Windchill object types. The schema consists of various object and attribute tables as well as internal tables used to track the migration status, pre-load validation data, and transformed, normalized data that represents the state of the data that is ready to be sent to the loaders.

It is recommended that the staging schema be created on a different, separate database for performance issues. The database user should be unique for migration activities.